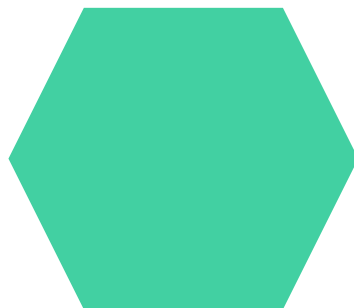
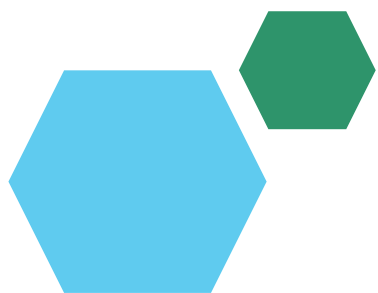




Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

- Agricultural performance is shaped by seasonal variation in rainfall, temperature, soil conditions and resource availability.
- Raw farm-level data does not clearly reveal how yield, cost and profit differ between the Kharif, Rabi and Zaid seasons.
- Problem: analyze seasonal agriculture data to uncover meaningful patterns, trends and relationships in performance across seasons.
- Goal: support evidence-based, season-specific agricultural planning and decision-making.



Project Description

Dataset: 4,000 farm-level records spanning 8 Indian states, 8 crops and 3 cropping seasons — Kharif, Rabi and Zaid.

Captures environmental conditions (rainfall, temperature, humidity, soil), resource usage (water, fertilizer, pesticide), and economic outcomes (cost, revenue, profit).

Objective: clean and explore the data, identify seasonal patterns and relationships, and derive evidence-based recommendations for seasonal agricultural planning.

Approach: exploratory data analysis, statistical testing (ANOVA) and visualization using Python, documented end-to-end in a Jupyter Notebook.

WHO ARE THE END USERS?

- **Farmers & Agronomists**

Plan crop selection, irrigation and inputs season-by-season using evidence rather than guesswork.

- **Agricultural Policy Makers**

Design seasonal advisories, subsidies and irrigation programs backed by real performance data.

- **Agribusiness & Supply Chains**

Anticipate seasonal shifts in production volumes and pricing to plan procurement and logistics.

- **Researchers & Students**

Use the findings as a reference case study on data-driven seasonal agricultural analysis.

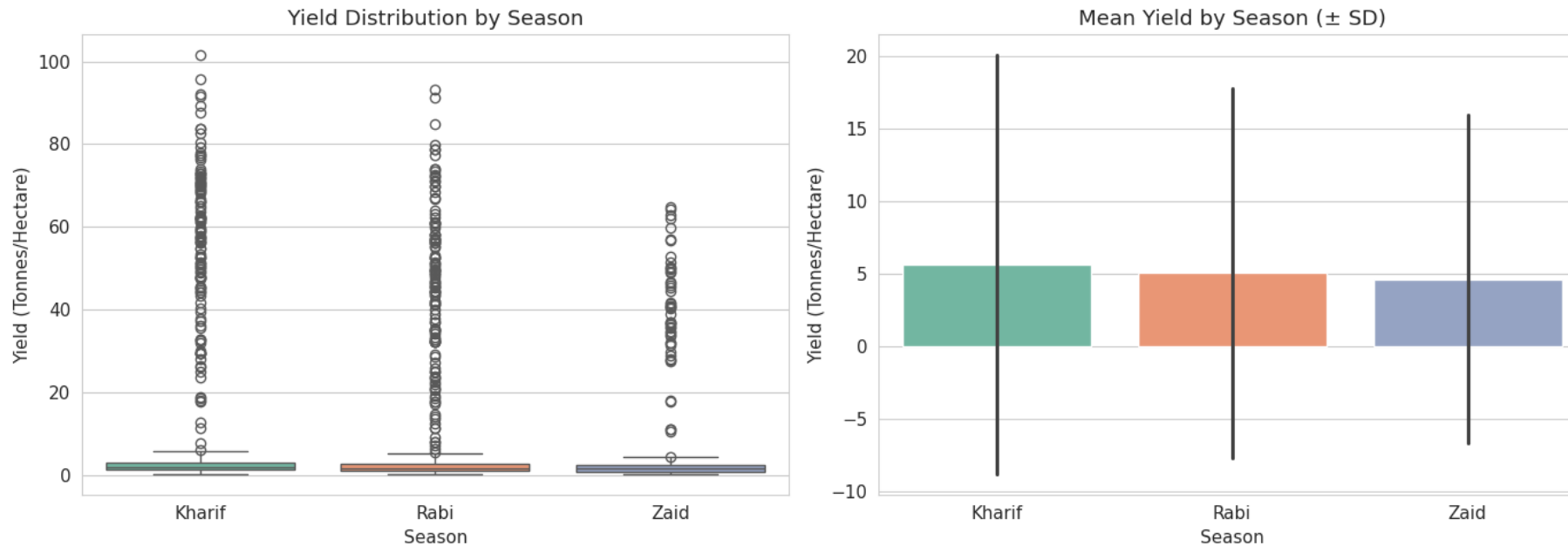
Technology Used

- **Python** — Core programming language for the entire analysis
- **Pandas & NumPy** — Data cleaning, transformation and numerical computation
- **Matplotlib & Seaborn** — Statistical charts, boxplots, heatmaps and trend visualizations
- **SciPy** — ANOVA and correlation testing for statistical significance
- **Jupyter Notebook** — End-to-end documentation of the analysis workflow



RESULTS

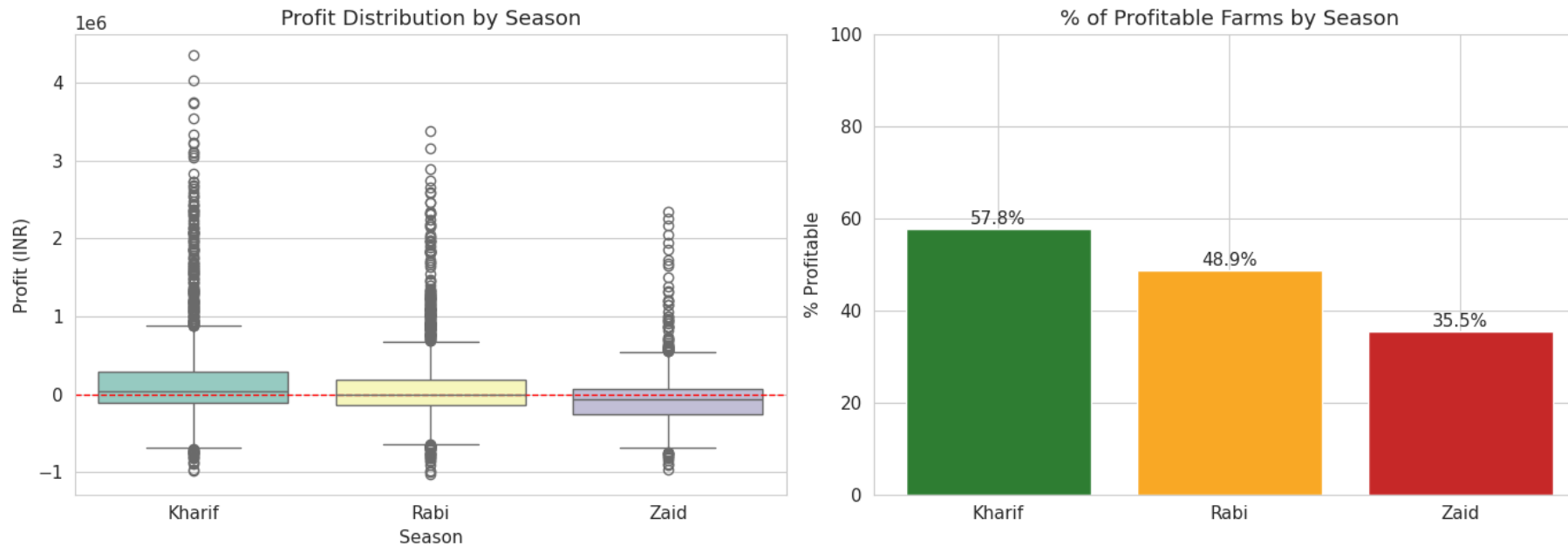
Yield Performance by Season



Kharif records the highest average yield (5.64 t/ha), aided by monsoon rainfall; Zaid is lowest (4.67 t/ha). A one-way ANOVA confirms this difference is statistically significant ($p < 0.05$).

RESULTS

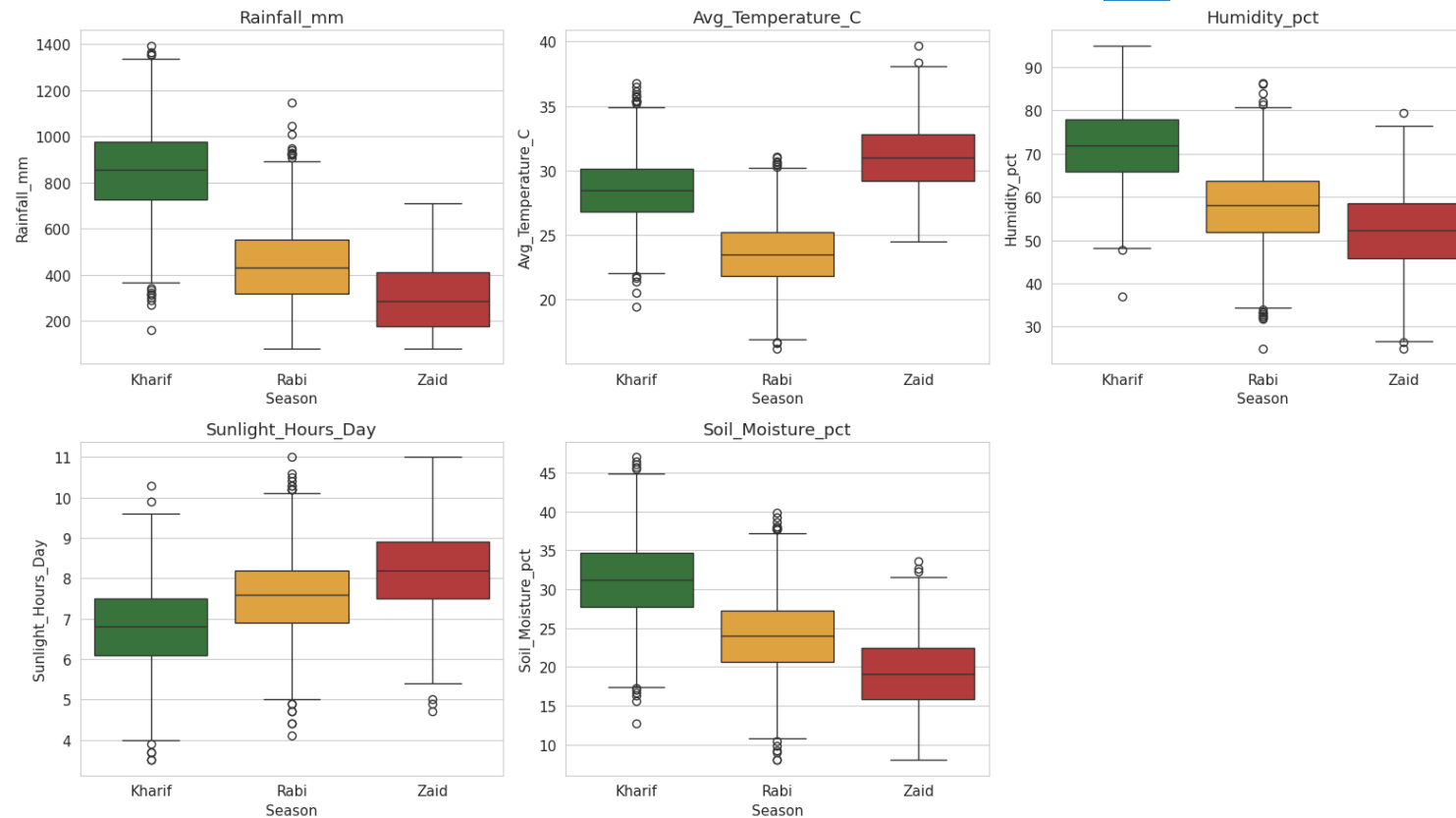
Economic Performance by Season



Kharif is the most profitable season on average (₹1,78,915; 57.8% of farms profitable). Zaid runs a net loss on average (-₹24,805) with only 35.5% of farms profitable.

RESULTS

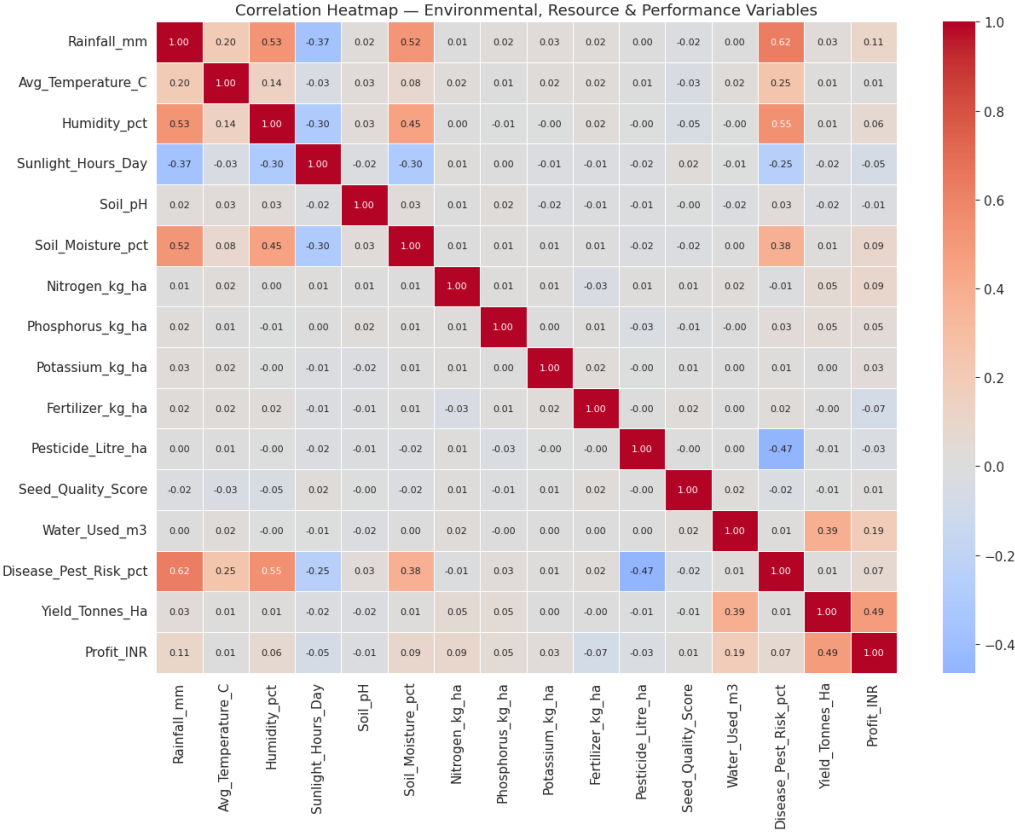
Environmental Conditions by Season



Kharif brings high rainfall and humidity from the monsoon; Zaid is hottest with the least rainfall and soil moisture – the primary driver of the seasonal yield and profit gap.

RESULTS

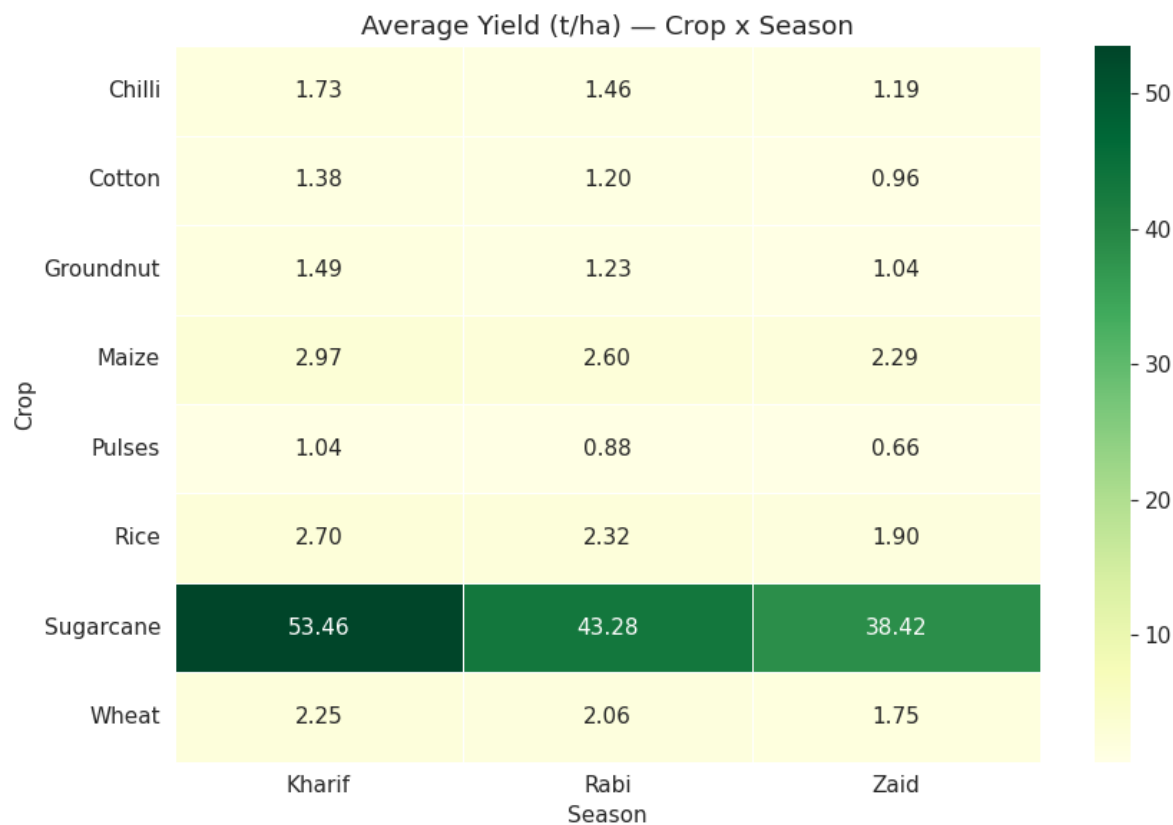
Relationships Between Key Variables



Yield correlates most strongly with water usage ($r = 0.39$); disease/pest risk rises sharply with rainfall and humidity ($r \approx 0.6$), linking wetter seasons to higher crop risk.

RESULTS

Crop Performance Across Seasons



Sugarcane shows the largest seasonal swing in yield, while most states also perform best in Kharif – confirming the seasonal effect holds consistently across crops and regions.

Future scope

- Incorporate multi-year data to separate genuine seasonal effects from single-year weather anomalies.
- Build predictive models (machine learning) to forecast yield and profit by season ahead of planting.
- Develop an interactive dashboard for real-time, season-wise monitoring of farm performance.
- Extend the analysis to district-level micro-climate and soil-type factors for finer-grained insight.
- Integrate market price forecasting with seasonal supply patterns to guide sell-timing decisions.

GitHub Link



<https://github.com/SriLalitha-CSE/Seasonal-Agriculture-Performance-Analysis-.git>



VOIS Course completion certificate -

Course name- Data Visualization





Thank you

